Cluster Robust Variance Estimation

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1 Introduction

In costructing the confidence intervals of the OLS estimator $\hat{\beta}_{\text{ols}}$, it is the asymptotic distribution of

$$\frac{1}{\sqrt{N}} \sum_{i=1}^{N} X_i u_i$$

that determines the estimator's asymptotic distribution. Assuming that $E[X_i u_i] = 0$, the Lindernberg-Levy CLT apply:

$$\frac{1}{\sqrt{N}} \sum_{i=1}^{N} X_i u_i \xrightarrow{d} \mathcal{N}\left(0, \frac{1}{N} \sum_i \mathrm{Var}\ (X_i u_i)\right)$$

But this hold true only if the individual observations are all independent, in fact summing 2 random variables X_1 and X_2 we get another random variable (call it Z) with expected value equal to $E[X_1] + E[X_2]$ and variance equal to $\text{Var}\ (X_1) + \text{Var}\ (X_2) + 2 \cdot \text{cov}\ (X_1, X_2)$. Summing N random variables $(X_1, ..., X_n)$, assume $E[X_i = 0]$, we get a new random variable $Z_N = \sum_{i=1}^{N} X_i$:

$$Z_N \sim \left(0, \sum_{i=1}^{N} \text{Var}\ (X_i) + 2 \sum_{i\neq}^{N} \sum_{j}^{N} \text{cov}\ \left(X_i, X_j\right)\right)$$

if all the X_is are **independent** all the joint moments (and so also the convariancers) are equal to zero and, dividing by $\sqrt{N}$:

$$\frac{1}{\sqrt{N}}(Z_N) \sim \left(0, \frac{1}{N} \sum_{i=1}^{N} \text{Var}\ (X_i)\right)$$

So the variance of Z_N scaled by $\frac{1}{\sqrt{N}}$ is the average variance of all the X_is. But here comes the problem, if our data are clustered we do not have that all the observation are independent, but rather we have that the observation of all the individuals inside a cluster are correlated, so we would have that

$$\frac{1}{\sqrt{N}} \sum_{i=1}^{N} X_i u_i \xrightarrow{d} \mathcal{N}\left(0, \frac{1}{N} \sum_{i=1}^{N} \text{Var}\ (X_i u_i) + \frac{2}{N} \sum_{i\neq}^{N} \sum_{j}^{N} \text{cov}\ \left(X_i u_i, X_j u_j\right)\right)$$

The variance estimator that is most commonly used, the White variance estimator assumes that all the covariances in the expression above are 0, and so estimates only $\frac{1}{N} \sum_{i=1}^{N} \text{Var}\ (X_i u_i)$, so it would not consistently estimate the asymptotic variance.

1.1 Why we estimate $\text{Var}\ \left(\hat{\beta}\right)$

To follow what comes next, it is essential to clarify **what we are actually estimating** when we estimate the variance of
$\hat{\beta}_{\text{ols}}$. After all, $\hat{\beta}_{\text{ols}}$ is just a number—so what does it mean to speak of its variance?
Is it a random variable? Does it have a distribution? Yes. The distribution we refer to is the distribution of $\hat{\beta}_{\text{ols}}$ **across repeated samples**, i.e., the values it would take if we changed the individuals drawn from the population. Since $\hat{\beta}_{\text{ols}}$ is a function of random variables (X_i and u_i), if we knew their full population distributions, we would know the exact value of β. But because we only observe a sample, our estimate varies from sample to sample. In this sense, $\hat{\beta}_{\text{ols}}$ *is* a random variable and therefore has a distribution.

1.2 What are clustered data?

Let's take a step back and ask ourselves what does it mean that our data are clustered? The defining feature of a cluster is that individuals within it exhibit some form of **unobserved** correlation.
Suppose our sample contains N individuals, each with an observed trait X. We can represent these as $X_{ig} = \left(X_{11}, ..., X_{N_1 1}, X_{12}, ..., X_{N_2 2}, ..., X_{N_G G}\right)$, where i indexes individuals within a cluster and g indexes the cluster.

We can think of all these X_{ig} as being drawn from the same population distribution, so they are **identically distributed**. Let us assume this distribution has mean 0.

However, individuals belonging to the same cluster are **not independent**. Their joint distribution is not simply the product of their marginals, their covariances may not be zero.
Saying that observations within a cluster are correlated means that knowing the value of one individual gives information about another individual in the same cluster. If individuals i and j belong to the same cluster g, then:

$$P\left(X_i > E[X] \mid X_j > E[X], i, j \in g\right) > P(X_i > E[X])$$

and similarly:

$$P\left(X_i < E[X] \mid X_j < E[X], i, j \in g\right) > P(X_i < E[X]).$$

outcomes inside the same cluster tend to move together relative to the overall mean—they are positively associated. If we focus on a single cluster g, we can view its observations as:

$$X_g = \{X_i \mid \text{ individual } \ i \in \text{ cluster } \ g\}$$

Each cluster is a subpopulation with its own distribution, and the overall population is the mixture of all G cluster distributions.

1.3 General setup

In this section, we lay out the general setup that will guide the rest of the thesis. We assume that there are heterogeneous effect across clusters, this means that the true parameter β is different in every cluster.

$$y_{ig} = \alpha + \beta_g X_{ig} + u_{ig}$$

We model u_{ig} has having a cluster component and an individual-and-cluster component:

$$u_{ig} = c_g + v_{ig}$$

Where $v_{ig} \sim (0, \sigma_v^2)$, $c_g \sim (0, \sigma_c^2)$ but of course c_g varies only at the cluster level. In this way the expected value of the error given that individual i is in cluster g is:

$$\mathbb{E}_{\mathbb{g}}\left(u_{ig}\right) = c_g$$

But as long as we have an intercept that varies for each cluster we can assume w.l.o.g that $\mathbb{E}\left[u_{ig}\right] = 0$.
We will call β_{APE} the average between cluster of the various β_g.
If we estimate β separately across G clusters, we obtain a vector of cluster-specific estimates $\left\{ \hat{\beta}_g \right\}_{g=1}^G$.

$$\widehat{\beta}_g = \beta_g + \left(\sum_{i \in g} X_i^2\right)^{-1} \left(\sum_{i \in g} X_i u_i\right)$$

And:

$$\hat{\beta}_{\mathrm{g}} \overset{a}{\sim} \mathcal{N}\left(\beta_g, \frac{1}{N_g}\left(\sum_{i \in g} X_i^2\right)^{-1} B\left(\sum_{i \in g} X_i^2\right)^{-1}\right)$$

where B_g is:

$$B_g = \mathrm{Var}\,\left(X_{ig} u_{ig}\right)$$

If for each cluster we could consistently estimate its asymptotic variance, we could take the average of these cluster-level estimates to obtain a consistent estimate of the asymptotic variance of the overall $\hat{\beta}_{\text{pols}}$, since a linear combination of normal random variables is also normal. But first we have to ask ourself wether each of the estimated parameters is consistent.

2 Assumption we make

We must begin by defining precisely the assumptions that we are willing make about the population.

The first choice is whether we treat the number of clusters as fixed and fully observed in every sample—for example, U.S. states (G = 50) when estimating the effect of a federal policy. Alternatively, we may assume an increasing number of clusters, where the specific clusters included can vary from sample to sample. In what follows, we discuss both cases.

Fixed amount of clusters

We can think of the data generating process (DGP) as having 2 stages:

- in the first stage all the population parameters are drawn. This include the vector of length G with all the β_g, and the cluster specific distributions for X_g and u_g. This stage only happens one, all the parameters from this stage are fixed.

- The second stage is a draw from the cluster distributions of (X, β, u) of the individual observations that will make our sample. This stage re-happens every time we take a sample from the population, this is the source of randomness.

Infinite amount of clusters

If the amount of clusters that we possibly have is not finite the first sampling stage reoccurs every time we sample. So there are two sources of randomness:

1. which clusters we draw from the possibly infinite population of clusters and,
2. which individuals we draw from those clusters.

2.1 Endogeneity

Then we must decide wether the sample we are working with presents cluster-level endogenity, i.e. if $\mathbb{E}[X_i u_i \mid i \in g] \neq 0$. In order for $\hat{\beta}_{\text{ols}}$ to be consistent we need exogenity, we need the regressors to be uncorrelated with the errors, but it may be that this requirement is met at the population level but not in all clusters, it may be that some clusters present correlation between the cluster specific portion of the errors and the regressors(A. C. Cameron, D. L. Miller, and others [1]), and that this correlation averages out across clusters. This would result in a biased estimation of the G $\widehat{\beta_g}$, but in a consistent estimation of the average partial effect.

We can think of it as if there is an unobserved variables, that has mean 0 in the population, but not in each cluster, and that is correlated with both the cluster regressor and the cluster error, making them correlated at the cluster level but averaging this correlation to 0 in the whole population.

3 Simulating a dataset

We will simulate data in two ways, in the first one we will make so that the expected value of the score is 0 in all clusters, in the second one the expected value of the score will be 0 only at the population level.

First we draw a random vector $\boldsymbol{\beta_g}$ from a normal distribution with mean β_{APE} and variance σ^2_β.

```
include("../src/ClusterSimulation.jl")
G = 50 #Number of clusters
Ng = 500 # Number of individuals per cluster

β = rand(Normal(2,4),G)
```

```
50-element Vector{Float64}:
  5.836607592105661
  6.028491822107145
  2.2024477962904405
 -6.970077305003979
  4.582936008825365
  5.131757313322424
  3.0620813707807084
  0.6835634409254991
 -0.20590227250233584
  2.1713482441880614
 -0.23153926712758066
 -1.8372907977459785
 -0.4879562564615254
  ⋮
  3.815608093165032
 -1.4189487804570962
  2.94196113519401
  0.8563519802177624
  3.473964403615745
 -1.340377019834305
 -1.6552746967106864
 -1.103399654296771
 -2.266427821957212
 -2.784052738811428
  6.122819295819502
  0.6821664963023804
```

We then simulate our dataset in two steps, according to the following model:

$$y_{ig} = \beta_g x_{ig} + (c_g + v_{ig})$$

In the first step we simulate the population parameters, and in the second we draw from them.

3.1 Description of the Two Steps

```
pop = hePopulation(G, β, randn(G))
```

```
Population with 50 clusters and with heterogeneous effect
```

This line creates a heterogeneous-effects population.
- **`G`** is the number of clusters.
- **`β`** is the vector of betas.
- **`randn(G)`** produces the vector $c_g \sim \mathcal{N}(0, 1)$.

The constructor returns a population struct that stores the cluster-level parameters used to simulate data.

```
sam = sample(pop, Ng)
```

```
Cluster sample struct
```

The line above generates a sample from the population.
- **`Ng`** is the number of observations per cluster.

The function draws data for all `G` clusters, each containing `Ng` observations, and returns a sample struct built according to the population's parameters.
I can then estimate $\hat{\beta}_{\text{g}}$ in each cluster and then broadcast it to all clusters in our sample, getting G estimates.
Each of these

3.2 Endogenous clusters

The second way in which we'll simulate our data is as follow.
We want to simulate data such that:

$$\mathbb{E}\left[X_{ig} u_{ig} \mid i \in \text{cluster} \quad g\right] = \mu_g$$

for each cluster g, and

$$\mathbb{E}\left[X_{ig} u_{ig}\right] = \mathbb{E}\left[\ \mathbb{E}\left[X_{ig} u_{ig} \mid i \in \text{cluster} \quad g\right]\ \right] = \mathbb{E}\left[\mu_g\right] = 0$$

Across all clusters, the overall expectation is zero because μ_g has mean zero. We first draw from a normal distribution $\mathcal{N}\left(0, \sigma_\mu^2\right)$ a vector $\boldsymbol{\mu}$ of length G, with all the cluster's mean μ_g. Note that in doing so we are taking a sample of size G from a theoretically infinite population of clusters.
Then we:

1. Draw two i.i.d. standard normal variables:

$$Z_1 \sim \mathcal{N}(0, 1), \quad Z_2 \sim \mathcal{N}(0, 1)$$

Z_1 and Z_2 are zero-mean and independent. The covariance between X and u will come entirely from the shared Z_1 component.

2. Construct within each cluster g:

$$X_g = s_g\, Z_1$$

$$u_g = a_g\, Z_1 + b_g\, Z_2$$

a_g controls the strength of the correlation within the cluster, b_g let u have positive variance.

$$a_g = \frac{\mu_g}{s_g}, \qquad b_g = \sqrt{s_g^2 - a_g^2}.$$

To make sure that b_g exist and is positive we need $s_g^4 \geq \mu_g^2$. So we choose:

$$s_g^2 = |\mu_g| + v_0 \quad \text{with} \quad v_0 > 0$$

So that:

$$\mathbb{E}[X \mid g] = \mathbb{E}[u \mid g] = 0$$

and

$$\mathbb{E}[Xu \mid g] = \text{Cov }(X, u \mid g) = s_g\, a_g = \mu_g.$$

We then compute Y_{ig} choosing a value for β.

$$Y_{ig} = X_{ig}\beta + u_{ig}$$

Let us now check that the expected value of the score is 0 in the whole population but not in each clusters.

```
beta2 = 2
μg = rand(Normal(0,4),500)
endoPop = clPopulation(500, μg, beta2)
endoSam = sample(endoPop,100)
S = checkScore(endoSam)
```

```
500-element Vector{Float64}:
  3.4286260488219096
  1.4907942107449532
  0.3755424060837393
  0.44410518482689426
  0.15520683888091866
  2.4710581845762007
 -2.8712090095692857
  4.374572168571519
  2.27230196013421
 -0.03269188757699749
```

```
-0.2080193772735602
 2.023061880829678
-1.6151278675476297
 ⋮
 1.9365597292079755
-0.6154513339050607
-1.697648214717339
 0.11190736200461342
-0.8283490124273908
 2.122763466885348
 0.37790635005105977
-1.0900783605117177
 0.7856294705430605
 0.5261065720811551
-1.0146323126347425
-0.4214799505609875
```

The above is the score in each cluster, now let's check te mean and variance.

```
mean(S), var(S)
```

```
(-0.030777007013596876, 4.407941443153037)
```

We can see that the mean is close to 0 and the variance is close to the expected variance of μ_g.

4 Consistency

We can discuss two way in which our estimator can be consistent, the first one is that:

$$\widehat{\boldsymbol{\beta}_{\boldsymbol{g}}} \xrightarrow{p} \boldsymbol{\beta}_{\boldsymbol{g}}$$

meaning that each of the G estimated parameters is consistent for its counterpart in the population. This would require that the expected value of the score is 0 in all clusters, i.e. no cluster-level endogeneity. The second way in which our estimator may be consistent is that:

$$\hat{\beta}_{\text{pols}} \xrightarrow{p} \beta_{\text{APE}}$$

For this to be the case we only need population exogeneity.

4.1 Montecarlo without endogeneity

We will now use a Montecarlo method to test wether we can get a consistent estimate of the G parameters if there is no cluster endogeneity. We will estimate the following model:

$$\mathbf{y} = \mathbf{X}_g \hat{\boldsymbol{\beta}}_{Ng} + \mathbf{u}_g$$

Where $\hat{\beta}_g = \begin{bmatrix} \alpha_g \\ \beta_g \end{bmatrix}$. Adding a cluster-specific intercept lets the model absorb cluster-level fixed effects in the residuals.

```
histogram(
    permutedims(stack(bols.(sam.allclusters)))[:,2] - β,
    bins = 15
    )
```

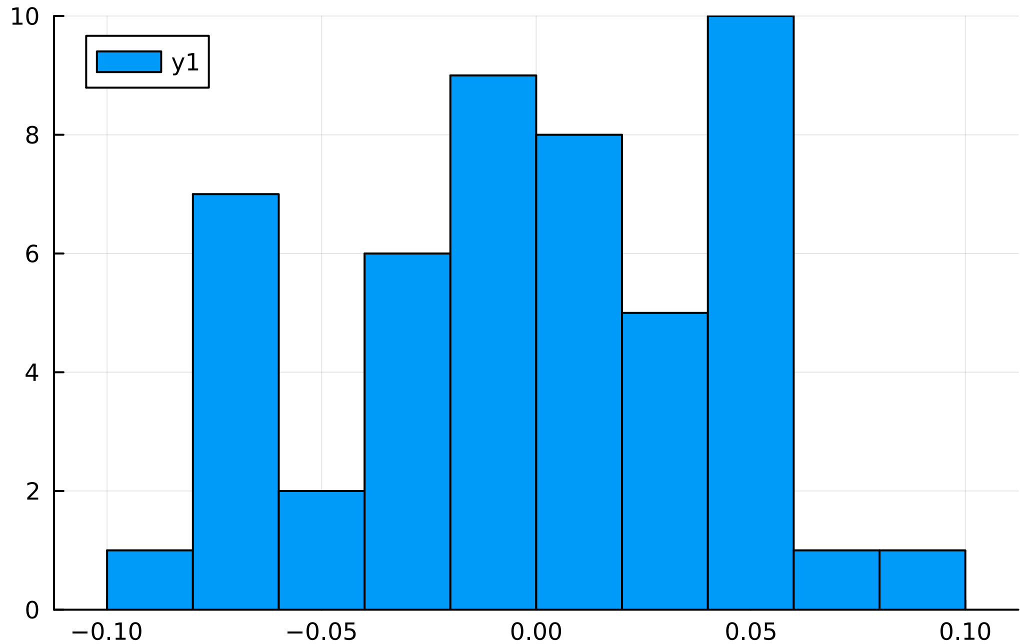


Figure 1: HIstogram of the difference between the estimated parameters and the true one in all clusters

In Figure 1 we can see a plot of the difference between the estimated value for β_g and the true value, it is clear from the graph that all the estimated parameters are consistent for their population counterpart. Now we'll run a Montecarlo simulation, with 100 iterations, we will simulate for each iteration 50 clusters of 100 people and compute $\hat{\beta}_g$ for each cluster. We will then compute the mean across simulation of the difference between the estimated β_g and the true paramenter.

```
betahatDiff = montecarloClusterWise(pop,100,100) .- β
histogram(mean(betahatDiff, dims = 2), bins =15)
```

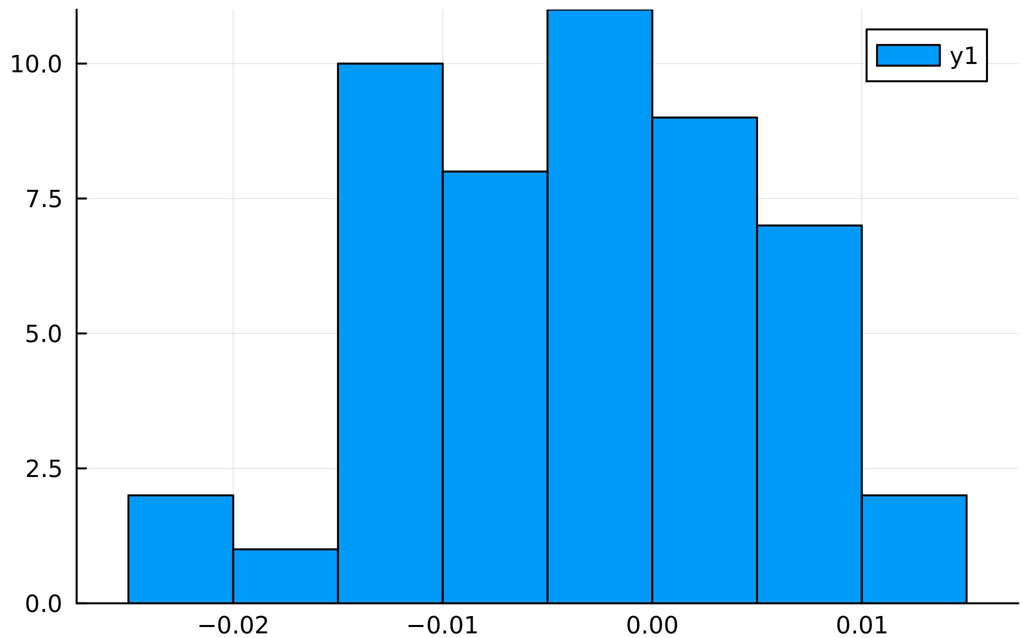


Figure 2: Histogram of the mean difference between estimated and true parameters across simulation in the montecarlo experiment

As can be clearly seen from Figure 2 there is consistency across all clusters.

4.2 APE

Since β_{APE} is simply the average of all the β_g, if each β_g can be consistently estimated, then β_{APE} can also be consistently estimated. But what happens if there is cluster-level endogeneity?
In that case, consistency is still achievable—as long as endogeneity does not arise at the population level. The key point is that estimating $\hat{\beta}_{\text{pols}}$ is equivalent to averaging the $\hat{\beta}_g$ across clusters. For each cluster we have

$$\widehat{\beta_g} \xrightarrow{p} \beta_g + \frac{\sum_{i \in g} S_i}{\sum_{i \in g} \mathbf{x_{i'}} \mathbf{x_i}}$$

where $S_i = X_i u_i$ is the score. Averaging this over all G clusters gives:

$$\hat{\beta}_{\text{pols}} = \frac{1}{G} \sum_g \widehat{\beta_g} \xrightarrow{p} \beta_{\text{APE}} + \frac{\sum_g \sum_{i \in g} S_i}{\sum_g \sum_{i \in g} \mathbf{x_{i'}} \mathbf{x_i}}$$

As long as we observe all clusters, the numerator of the second term is equal to 0, and the estimator is therefore consistent.

4.3 Consistency as $N_g \xrightarrow{p} \infty$

We want to test wether our estimate of the average partial effect gets better if the number of people that we draw in each cluster increases. In order to do so we ran a series of montecarlo simulation increasing N_g.

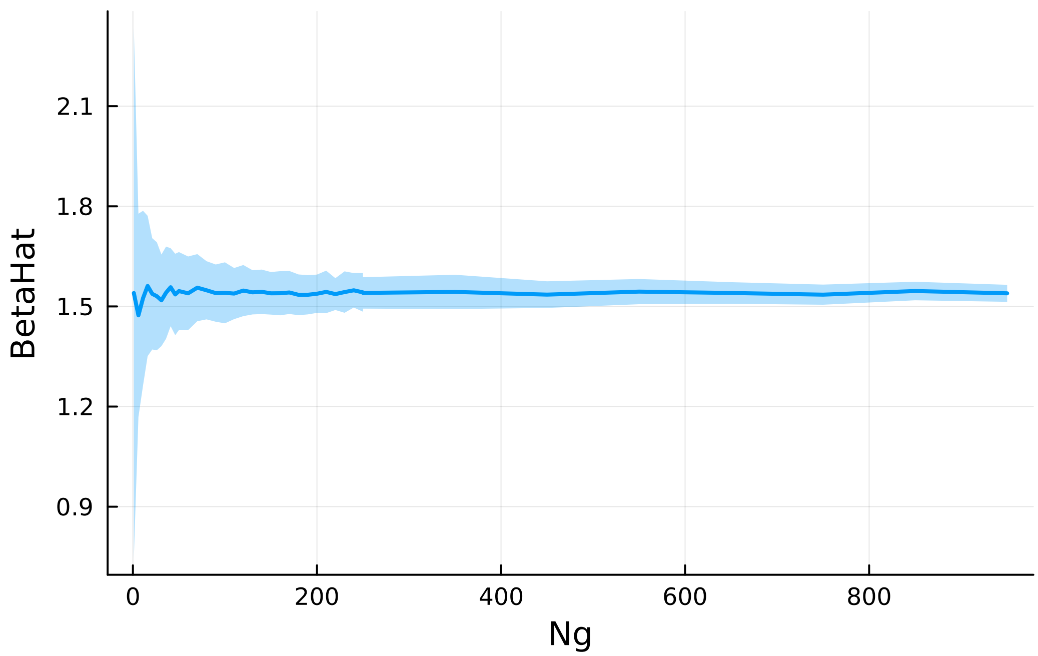


Figure 3: Consistency of betahat as Ng

What we see in Figure 3 is the result of the montecarlo simulation, the line is the average estimates for each N_g, while the shaded area is the standard deviation, we can clearly see from the graph that as the number of observations inside a cluster grows $\hat{\beta}_{\text{ols}} \xrightarrow{p} \beta_{\text{APE}}$.

5 Variance Estimation

In each cluster we have that

$$\widehat{\beta_g} \xrightarrow{p} \beta_g + \frac{\sum_{i \in g} S_i}{\sum_{i \in g} \mathbf{x_{i'}} \mathbf{x_i}}$$

Where S_i is the score $X_i u_i$.

$$\left(\widehat{\beta_g} - \beta_g\right) \xrightarrow{d} \mathcal{N}(0, V_g)$$

$$V_g = \sum_{i \in g} \mathbf{x_{i'}} \mathbf{x_i} \left(\text{Var}\left(\sum_{i \in g} S_i\right)\right) \sum_{i \in g} \mathbf{x_{i'}} \mathbf{x_i}$$

Since inside each cluster observations can be considered i.i.d.

$$\mathrm{Var}\left(\sum_{i\in g} S_i\right) = \sum_{i\in g} \mathrm{Var}\ (Si)$$

The variance is then

$$\mathbb{E}\left(S_{ig}^2\right) - \left[\mathbb{E}\left(S_{ig}\right)^2\right]$$

Now if we want to estimate β_{APE} we note that it is equal to the average across G of the various β_g, and since the vaious $\widehat{\beta}_g$ are independent, the variance of their mean will be the mean of their variances.

$$\sum_g \left(\widehat{\beta}_g - \beta_g\right) = \hat{\beta}_{\text{pols}} - \beta_{\text{APE}} \overset{a}{\sim} \mathcal{N}\left(0, \frac{1}{G}\sum_g \mathrm{Var}\ \left(\widehat{\beta}_g\right)\right)$$

As noted before if we have enough clusters:

$$\sum_g \sum_{i\in g} S_i \xrightarrow{p} \mathbb{E}_g\left(S_{ig}\right) = 0$$

So we get a consistent estimate even if there is cluster level endogeneity. But this not apply to the variance. That is because

$$\sum_g \sum_{i\in g} S_i^2 \xrightarrow{p} \mathbb{E}_g\left(S_{ig}\right) \neq 0$$

So that there is not averaging out across clusters that make the second term disappear. This means that the asymptotic variance of $\hat{\beta}_{\text{pols}}$ will be different depending on wether there is cluster-level endogeneity or not, and the difference will be equals to:

$$V_{\text{endo}} - V_{\text{exo}} = \frac{\mathbb{E}_g\left(\mu_g^2\right)}{\sum_i \mathbf{x_{i'}x_i}}$$

5.1 All cluster observed and no endogeneity

So if there is no endogeneity i can consistently estimate the AVar of each $\widehat{\beta}_g$ using White variance estimator, let's test this using the montecarlo method.

```
meanbeta, sdbeta = montecarlo(pop,1000,Ng)
D = Normal(meanbeta,sdbeta)

x = range(meanbeta - 4*sdbeta,meanbeta + 4*sdbeta; length = 1000)
j = pdf.(D,x)

plot(x,j, linewidth=2, label="Montecarlo")

betahat = bols(sam)[2]
sigma =   sqrt(whiteAvar(sam)[2,2])
```

```
d = Normal(betahat, sigma)

y = pdf.(d, x)

plot!(x, y, linewidth= 2, label = "White")
```

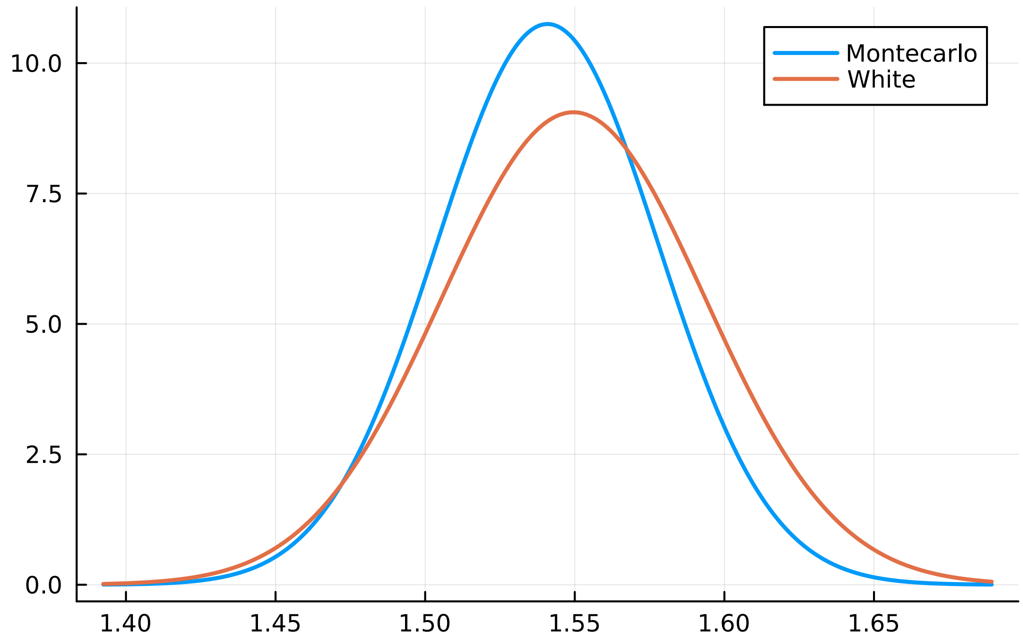
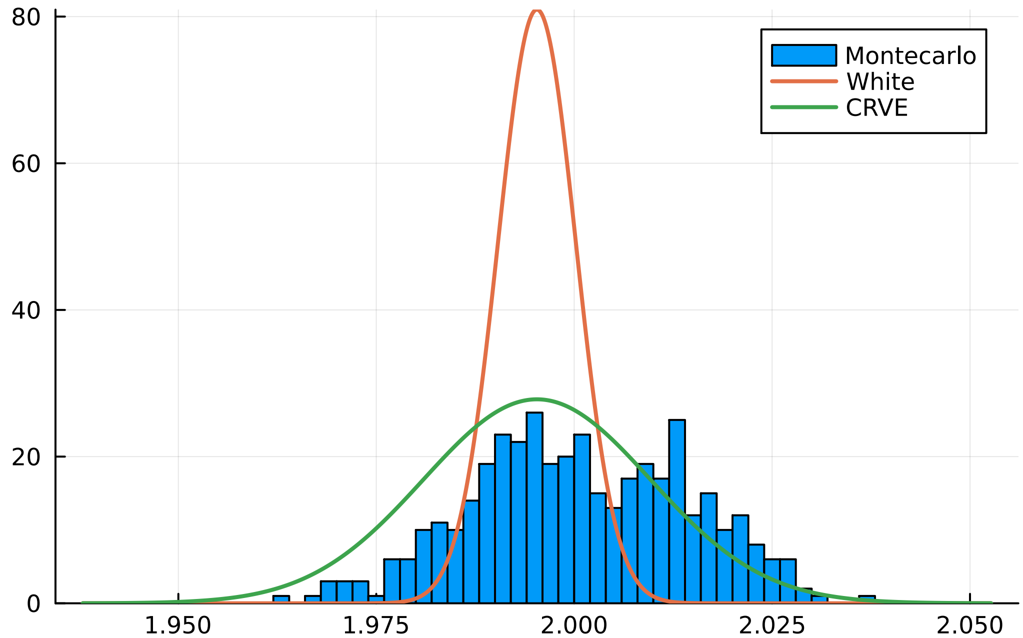


Figure 4: Consistency of White Variance estimator without cluster level endogeneity

Figure 4

```
betahat = bols(endoSam)[2]
white = sqrt(whiteAvar(endoSam)[2,2])
crve = sqrt(CRVE(endoSam)[2,2])
histogram(montecarlo(endoPop,400,Ng,true,true), bins=50, label="Montecarlo")
d = Normal(betahat, white)
D = Normal(betahat, crve)

x = range(betahat - 4 * crve, betahat + 4 * crve; length = 1000)
y = pdf.(d,x)
Y = pdf.(D,x)
plot!(x,y, linewidth =2, label = "White")
plot!(x,Y, linewidth =2, label = "CRVE")
```



6 CRVE Matrix

6.1 Notation

We use the same notation as in [2]:

$$\mathbf{X_{ig}} = \left[x_{1ig},\ x_{2ig},\ \dots\ x_{kig}\right]$$

$$\mathbf{X_g} = \begin{bmatrix} \mathbf{X_{ig}} \\ \mathbf{X_{jg}} \\ \dots \\ \mathbf{X_{N_g g}} \end{bmatrix} = \begin{bmatrix} x_{1ig}, & x_{2ig}, & \dots & x_{kig} \\ x_{1ij}, & x_{2jg}, & \dots & x_{kjg} \\ \dots & & & \\ x_{1N_g g}, & x_{2N_g g}, & \dots & x_{kN_g g} \end{bmatrix}$$

$$
\mathbf{X} = \begin{bmatrix} \mathbf{X_{ig}} \\ \mathbf{X_{jg}} \\ \dots \\ \mathbf{X_{N_g g}} \\ \mathbf{X_{ih}} \\ \mathbf{X_{jh}} \\ \dots \\ \mathbf{X_{N_h h}} \\ \dots \\ \mathbf{X_{iG}} \\ \mathbf{X_{jG}} \\ \dots \\ \mathbf{X_{N_G G}} \end{bmatrix}
$$

–

The idea behind CRVE is that clusters are independent one from the other. Since inference requires independence across a big number of units, we base it on the number of clusters rather than the number of individuals. Assuming that $\mathbb{E}\left[\mathbf{X}_\mathbf{g}'\mathbf{u}_\mathbf{g}\right] = 0$, $\hat{\beta}_{\text{ols}}$ will be a consistent estimator of the population parameter β if $G \to \infty$ [3]. To understand this note that computing β:

$$
(X'X)^{-1}(X'y)
$$

is the same as doing:

$$
\hat{\boldsymbol{\beta}}_{\text{ols}} = \left(\sum_g^G \mathbf{X_g}'\mathbf{X_g}\right)^{-1}\left(\sum_g^G \mathbf{X_{g'}}\mathbf{y_g}\right)
$$

so

$$
\left(\hat{\boldsymbol{\beta}}_{\text{ols}} - \boldsymbol{\beta}\right) = \left(\frac{1}{G}\sum_g^G \mathbf{X_g}'\mathbf{X_g}\right)^{-1}\left(\frac{1}{G}\sum_g^G \mathbf{X_{g'}}\mathbf{u_g}\right)
$$

and applying the Weak Law of Large Numbers J. M. Wooldridge [3]

$$
\frac{1}{G}\sum_g^G \mathbf{X_{g'}}\mathbf{u_g} \xrightarrow{p} \mathbb{E}\left[\mathbf{X}_\mathbf{g}'\mathbf{u_g}\right] = 0
$$

so $\hat{\boldsymbol{\beta}}_{\text{ols}}$ will be consistent. Now let us derive it's asymptotic distribution:

$$
\sqrt{G}\left(\hat{\boldsymbol{\beta}}_{\text{ols}} - \boldsymbol{\beta}\right) = \left(\frac{1}{G}\sum_g^G \mathbf{X_g}'\mathbf{X_g}\right)^{-1}\left(\frac{1}{\sqrt{G}}\sum_g^G \mathbf{X_{g'}}\mathbf{u_g}\right)
$$

The first term is by assumption $A^{-1} + o_p(1)$, while the second is $\overset{a}{\sim} \mathcal{N}\left(0, \mathbb{E}\left[\mathbf{X_{g'}}\mathbf{u_g}\mathbf{u_{g'}}\mathbf{X_g}\right]\right)$ So

$$\sqrt{G}\left(\hat{\beta}_{\text{ols}}-\beta\right) \xrightarrow{d} \mathcal{N}\left(0, A^{-1}\mathbb{E}\left[\mathbf{X}_{\mathbf{g}'}\mathbf{u}_{\mathbf{g}}\mathbf{u}_{\mathbf{g}'}\mathbf{X}_{\mathbf{g}}\right]A^{-1}\right)$$

6.2 Formula

$$\hat{V}_{\text{clu}}\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{B}(\mathbf{X}'\mathbf{X})^{-1} = (\mathbf{X}'\mathbf{X})^{-1}\left(\sum_{g=1}^{G}\mathbf{X}_g{}'\hat{\mathbf{u}}_g\hat{\mathbf{u}}_g{}'\mathbf{X}_g\right)(\mathbf{X}'\mathbf{X})^{-1}$$

Define:

$$\mathbf{s}_{\mathbf{g}} \equiv \mathbf{X}_{\mathbf{g}}{}'\hat{\mathbf{u}}_{\mathbf{g}} = \begin{bmatrix} X_{11g} & X_{12g} & \cdots & X_{1N_gg} \\ X_{21g} & X_{22g} & \cdots & X_{2N_gg} \\ \vdots & \vdots & \ddots & \vdots \\ X_{k1g} & X_{k2g} & \cdots & X_{kN_gg} \end{bmatrix} \begin{bmatrix} \hat{u}_{1g} \\ \hat{u}_{2g} \\ \vdots \\ \hat{u}_{N_gg} \end{bmatrix} = \begin{bmatrix} \sum_{i=1}^{N_g} X_{1ig}\,\hat{u}_{ig} \\ \sum_{i=1}^{N_g} X_{2ig}\,\hat{u}_{ig} \\ \vdots \\ \sum_{i=1}^{N_g} X_{kig}\,\hat{u}_{ig} \end{bmatrix}$$

Then:

$$\mathbf{s}_{\mathbf{g}}\mathbf{s}_{\mathbf{g}}{}' = \left(X_g{}'u_g\right)\left(X_g{}'u_g\right)' = X_{g'}u_gu_{g'}X_g$$

$$\mathbf{s}_{\mathbf{g}}\mathbf{s}_{\mathbf{g}}{}' = \begin{bmatrix} \sum_{i\in g}\sum_{j\in g} X_{1ig}X_{1jg}\hat{u}_{ig}\hat{u}_{jg} & \sum_{i\in g}\sum_{j\in g} X_{1ig}X_{2jg}\hat{u}_{ig}\hat{u}_{jg} & \cdots & \sum_{i\in g}\sum_{j\in g} X_{1ig}X_{kjg}\hat{u}_{ig}\hat{u}_{jg} \\ \sum_{i\in g}\sum_{j\in g} X_{2ig}X_{1jg}\hat{u}_{ig}\hat{u}_{jg} & \sum_{i\in g}\sum_{j\in g} X_{2ig}X_{2jg}\hat{u}_{ig}\hat{u}_{jg} & \cdots & \sum_{i\in g}\sum_{j\in g} X_{2ig}X_{kjg}\hat{u}_{ig}\hat{u}_{jg} \\ \vdots & \vdots & \ddots & \vdots \\ \sum_{i\in g}\sum_{j\in g} X_{kig}X_{1jg}\hat{u}_{ig}\hat{u}_{jg} & \sum_{i\in g}\sum_{j\in g} X_{kig}X_{2jg}\hat{u}_{ig}\hat{u}_{jg} & \cdots & \sum_{i\in g}\sum_{j\in g} X_{kig}X_{kjg}\hat{u}_{ig}\hat{u}_{jg} \end{bmatrix}$$

Then we sum this over all G clusters: \$\$ = \$\$

So what CRVE does is to compute for each cluster the outer product of $\hat{\mathbf{S}}_{\mathbf{g}} = \mathbf{X}_{\mathbf{g}'}\hat{\mathbf{u}}_{\mathbf{g}}$ with itself, and then average those matrices across clusters.
After pre and post-multiplication by $\left(\frac{1}{N}\sum_i \mathbf{x}_i\mathbf{x}_i'\right)^{-1}$, this converges to the **asymptotic variance-covariance matrix** of the OLS estimator $\hat{\beta}_{\text{ols}}$.

6.3 Convergence

We want to show that

$$\frac{1}{G}\sum_g \mathbf{X}_g'\,\hat{\mathbf{u}}_g\hat{\mathbf{u}}_g'\mathbf{X}_g \xrightarrow{p} \mathbb{E}\left[\mathbf{X}_g'\mathbf{u}_g\mathbf{u}_g'\mathbf{X}_g\right]$$

First let's substitute for $\hat{\mathbf{u}}_{\mathbf{g}}$:

$$\frac{1}{G}\sum_g \mathbf{X}_g'\left(\,\mathbf{y}_g - \mathbf{X}_g\hat{\beta}\,\right)\left(\,\mathbf{y}_g - \mathbf{X}_g\hat{\beta}\right)'\mathbf{X}_g$$

$$\frac{1}{G}\sum_g \mathbf{X}_g'\left(\,\mathbf{X}_g\beta + \mathbf{u}_g - \mathbf{X}_g\hat{\beta}_g\,\right)\left(\,\mathbf{X}_g\beta + \mathbf{u}_g - \mathbf{X}_g\hat{\beta}_g\right)'\mathbf{X}_g$$

We then factor terms in order to isolate $\left(\beta - \hat{\beta}\right)$

$$\frac{1}{G}\sum_g \mathbf{X}_g'\left[\mathbf{X}_g\ (\ \beta-\hat{\beta}_g\ )+\mathbf{u}_g\right]\left[\mathbf{X}_g\left(\beta-\hat{\beta}_g\right)+\mathbf{u}_g\right]'\mathbf{X}_g$$

As $\hat{\beta}\xrightarrow{p}\beta$, the extra terms vanish, leaving only $\mathbf{u}_g$.

$$\xrightarrow{p}\frac{1}{G}\sum_g \mathbf{X}_g'\mathbf{u}_g\mathbf{u}_g'\mathbf{X}_g\xrightarrow{p}\mathbb{E}\left[\mathbf{X}_g'\mathbf{u}_g\mathbf{u}_g'\mathbf{X}_g\right]$$

6.4 Difference from White

Let's look at the difference between the White covariance matrix and the CRVE. Imagine we have a cluster of 2 observations, the White inner matrix would like this:

$$B_{\mathrm{W}}=\begin{bmatrix}\widehat{u_1}^2 & 0\\ 0 & \widehat{u_2^2}\\ & \end{bmatrix}$$

While the CRVE would look like this:

$$B_{\mathrm{C}}=\begin{bmatrix}\hat{u}_1^2 & \hat{u}_1\hat{u}_2 & \hat{u}_1\hat{u}_3\\ \hat{u}_1\hat{u}_2 & \hat{u}_2^2 & \hat{u}_2\hat{u}_3\\ \hat{u}_1\hat{u}_3 & \hat{u}_2\hat{u}_3 & \hat{u}_3^2\end{bmatrix}$$

So when I then pre and post multiply by X:

$$X'B_{\mathrm{W}}X=x_1^2u_1^2+x_2^2u_2^2=\sum_i x_i^2u_i^2$$

$$X'B_{\mathrm{C}}X=x_1^2\hat{u}_1^2+x_2^2\hat{u}_2^2+x_3^2\hat{u}_3^2+2(x_1x_2\hat{u}_1\hat{u}_2+x_1x_3\hat{u}_1\hat{u}_3+x_2x_3\hat{u}_2\hat{u}_3)=X'B_{\mathrm{W}}X+2\sum_i\sum_{\neq j}x_ix_j\hat{u}_i\hat{u}_j$$

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